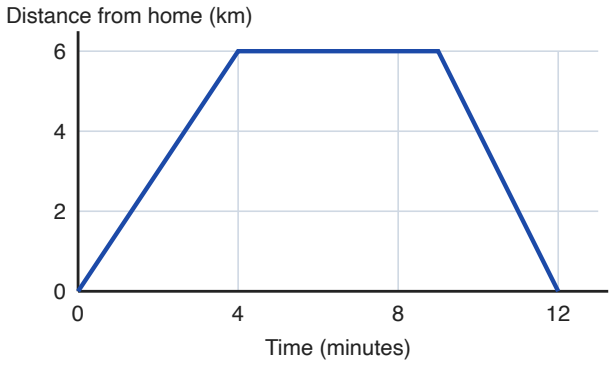
Graphs 3 — Distance–time graphs · Speed–time graphs

Calculator allowed. Squared paper for the drawing question. State units in every answer. Marks in brackets. ★ = harder question.

1 The travel graph shows Anjali's journey to a friend's house and back. [6]



She leaves at 0 min, arrives (6 km) at 4 min, stays until 9 min, and is home at 12 min.

a Calculate her outward speed in km/min.

b How long did she stay at her friend's house?

c Calculate her return speed in km/h.

a Gradient of the first section: $\frac{6}{4} = 1.5$ km/min.

b The flat part: $9 - 4 = 5$ minutes.

c Return: 6 km in 3 min $= 2$ km/min $= 2 \times 60 = 120$ km/h (the graph slopes down because she is coming back).

2 A drone accelerates steadily from rest to 15 m/s in 25 s, flies at that speed for 35 s, then slows steadily to rest in a further 40 s. [10]

a Draw the speed–time graph of the flight.

b Use the graph to find the drone's **(i)** initial acceleration, **(ii)** final acceleration, **(iii)** average speed over the whole 100 s flight.

a A trapezium: $(0, 0)$ up to $(25, 15)$, level to $(60, 15)$, down to $(100, 0)$.

b (i) Gradient $= \frac{15}{25} = 0.6$ m/s². **(ii)** $\frac{0 - 15}{40} = -0.375$ m/s² (a deceleration of 0.375 m/s²).

(iii) Distance = area under the graph:
$\frac{1}{2} \times 25 \times 15 + 35 \times 15 + \frac{1}{2} \times 40 \times 15 = 187.5 + 525 + 300 = 1012.5$ m. Average speed $= \frac{1012.5}{100} = 10.1$ m/s (3 s.f.).

3 The winner of a 400 m race finished in 50.0 seconds. [9]

a Calculate his average speed in m/s.

b Change your answer to km/h.

c The sketch of his speed–time graph shows him accelerating uniformly from rest for the first 6 seconds, then running at a constant maximum speed for the rest of the race. Calculate his maximum speed, correct to 3 s.f.

d Calculate his acceleration over the first 6 seconds, with units, correct to 2 s.f.

> **a** $\dfrac{400}{50} = 8$ m/s.
> **b** $8 \times 3600 \div 1000 = 28.8$ km/h.
> **c** Let the maximum speed be v. Distance = area: triangle $\frac{1}{2} \times 6 \times v$ plus rectangle $44 \times v$, so $3v + 44v = 47v = 400 \Rightarrow v = 8.51$ m/s.
> **d** $\dfrac{8.51}{6} = 1.4$ m/s² (2 s.f.).

4 A train leaves a station and accelerates uniformly for 2 minutes until it reaches 30 m/s, travels at 30 m/s for 10 minutes, then decelerates uniformly to rest in 90 seconds at the next station. Calculate the distance between the stations in kilometres. [5]

> Work in seconds: 120 s, 600 s, 90 s. Area under the speed–time graph:
> $\frac{1}{2} \times 120 \times 30 + 600 \times 30 + \frac{1}{2} \times 90 \times 30 = 1800 + 18\,000 + 1350 = 21\,150$ m
> $= 21.15$ km.

Total: 30 marks

Check: distance–time — gradient = speed, flat = stopped, down = returning · speed–time — gradient = acceleration, *area* = distance · convert minutes to seconds (or hours) before calculating · m/s → km/h: × 3.6.